Solved Exercises

Biology - Class 11

Punjab Curriculum and Textbook Board (PCTB)

Solved exercises of the PCTB Class 11 Biology textbook (English medium; eLearn.Punjab version, 455 pages; described on the Taleem360 detail page as the official edition for 2020 and onwards). Every chapter-end exercise is solved: fill in the blanks, True/False with corrections, short questions and extensive questions (Chapter 10 also has choose-the-correct-option items). Answers are written from the chapter text. Where the printed exercise has a numbering slip or a scrambled list, a key note at the end of the chapter records it. Source: Class11-Biology-taleem360.pdf (Taleem360 PCTB listing, complete book).

Chapter 1: Introduction (Exercise pp. 22-23)

Exercise - Fill in the blanks

(i) _____ is the study of organisms in relation to their environment.

Answer: **Ecology**

(ii) The study of organisms living in fresh water bodies like rivers, lakes etc. is called _____.

Answer: **Limnology**

(iii) _____ is the branch of biology which deals with the study of social behaviour and communal life of human beings.

Answer: **Sociobiology**

(iv) In the _____ body only six bio-elements account for 99% of the total mass.

Answer: **human**

(v) All living things and nonliving things are formed of simple units called _____.

Answer: **atoms**

(vi) Various organs in plants and various organ systems in animals are assembled together to form an _____.

Answer: **organism**

(vii) A _____ is a group of organisms of the same species located in the same place at the same time.

Answer: **population**

(viii) A _____ is based upon observations.

Answer: **hypothesis**

(ix) A hypothesis is a result of deductive reasoning or it can be the consequence of _____ reasoning.

Answer: **inductive**

Exercise - Write whether the statement is true or false (correct it if false)

(i) Penicillin was discovered by Edward Jenner from a fungus Penicillium.

Answer: False - Correction: Penicillin was discovered by Alexander Fleming from the fungus Penicillium notatum. Edward Jenner is known for smallpox vaccination.

(ii) Many diseases such as polio, whooping cough, measles, mumps etc. can be controlled by antibiotics.

Answer: False - Correction: These diseases are caused by viruses (and whooping cough by a bacterium but controlled by vaccination); antibiotics kill bacteria, not viruses. They are controlled by vaccination, not antibiotics.

(iii) Exposure to the small pox virus allows the body to develop immunity against cowpox virus.

Answer: False - Correction: It is the reverse: exposure to the mild cowpox virus allows the body to develop immunity against the smallpox virus. This was Jenner's discovery.

(iv) AIDS is caused by HIV and it spreads through sexual contacts, blood transfusion, by contaminated syringe or surgical instruments.

Answer: True

Exercise - Short questions

(i) What do you mean by hypothesis?

A hypothesis is a tentative statement, based on observations, that is put forward to explain a biological problem and can be tested by experiments. For example, 'malaria is caused by a germ carried by mosquitoes' was a hypothesis before it was proved. A hypothesis must be testable: experiments either support it or reject it.

(ii) How does law differ from theory?

A law is a statement that describes what happens, usually in a short, exact form, such as the law of segregation. A theory is a well-tested explanation of why something happens, built from many observations and experiments, such as the germ theory of disease. A law states a fact of nature; a theory explains it. A theory does not grow into a law.

(iii) What is deductive reasoning?

Deductive reasoning is the method of drawing a specific conclusion from a general rule or hypothesis. It moves from the general to the specific: if the hypothesis is true, then a particular result must follow, and an experiment is designed to test that prediction. For example, from 'all mosquitoes carry malaria' one deduces that a person bitten by a mosquito will get malaria, which can then be tested.

(iv) Define vaccination.

Vaccination is the introduction of a weakened, killed or harmless part of a disease-causing germ into the body so that the immune system learns to recognise it and makes antibodies and memory cells. When the real germ later enters the body, the immune system destroys it quickly, so the person does not fall ill. Jenner's cowpox vaccination against smallpox was the first example.

(v) Write a short note on cloning.

Cloning is the production of genetically identical copies of a cell, tissue or whole organism from a single parent, without sexual reproduction. In animals it is done by transferring the nucleus of a body cell into an egg cell whose own nucleus has been removed, and letting it develop. The first famous animal clone was Dolly the sheep. Cloning is used in research, medicine and agriculture, but it raises ethical questions.

Exercise - Extensive questions

(i) Define the following branches of biology: Molecular Biology, Microbiology, Marine Biology, Biotechnology.

Molecular Biology is the study of the structure and function of biological molecules, such as DNA, RNA and proteins, and how they control life. Microbiology is the study of microscopic organisms, such as bacteria, viruses and fungi. Marine Biology is the study of organisms living in the sea and oceans. Biotechnology is the use of living organisms and their systems to make useful products, such as medicines, vaccines and improved crops.

(ii) Discuss briefly phyletic lineage in biological organization.

Phyletic lineage means the evolutionary line of descent of a group of organisms: the sequence of ancestors and descendants that connects them through time. In biological organization it shows how simpler forms gave rise to more complex forms, for example how single cells led to many-celled organisms with tissues, organs and organ systems. The lineage of a group is studied through fossils and through similarities in structure, embryos and molecules.

(iii) Write notes on the following: (a) Living world in space and time (b) Population (c) Community.

(a) Living world in space and time: life exists almost everywhere on earth, from deep oceans to high mountains, and it has existed for about 3.5 billion years, changing through evolution. (b) Population: a group of organisms of the same species living in the same place at the same time, such as all the frogs in a pond. (c) Community: all the populations of different species living together and interacting in one area, such as all the plants, animals and microbes of a pond.

(iv) Explain the biological method for solving a biological problem. How do deductive and inductive reasoning play an important role in it?

The biological method has these steps: (1) Recognise a biological problem from observations. (2) Form a hypothesis to explain it. (3) Make deductions: if the hypothesis is true, certain results must follow. (4) Test the deductions by experiments with controls. (5) Record and analyse the results. (6) Accept, reject or modify the hypothesis, and report the conclusion. Inductive reasoning works from many particular observations to a general hypothesis, while deductive reasoning works from the general hypothesis to particular predictions that experiments can test. Both are needed: induction creates the idea,

deduction tests it.

(v) What is the role of the study of Biology in the welfare of mankind?

Biology improves human welfare in many ways: it gives better crops and livestock through breeding and biotechnology, so food increases; it discovers medicines, vaccines and antibiotics that control disease; it explains how the body works, helping doctors treat illness; it teaches how to protect the environment, soil, water and wildlife; and it supports industries based on fermentation, fisheries and forestry. In short, biology raises health, food, wealth and the quality of life.

Key notes (this chapter):

- Chapter 1 has no question 3 in print: the numbering jumps from Q.2 to Q.4.

Chapter 2: Biological Molecules (Exercise p. 55)

Exercise - Fill in the blanks

(i) The sum of all the chemical reactions taking place within a cell is called _____.

Answer: **metabolism**

(ii) _____ is the basic element of organic compounds.

Answer: **Carbon**

(iii) All the amino acids have an amino group and a carboxyl group attached to the same _____ atom.

Answer: **alpha carbon (central carbon)**

(iv) _____ is the most abundant carbohydrate in nature.

Answer: **Cellulose**

(v) Adenine and guanine are double ringed bases and are called _____.

Answer: **purines**

Exercise - Write whether the statement is true or false (correct it if false)

(i) A small proportion of water molecules are in ionized form.

Answer: True

(ii) The covalent bond among two monosaccharides is called a peptide bond.

Answer: False - Correction: The bond between two monosaccharides is a glycosidic bond. A peptide bond joins two amino acids.

(iii) Glycogen is also called plant starch.

Answer: False - Correction: Glycogen is animal starch, stored in liver and muscles. Plant starch is a mixture of amylose and amylopectin.

(iv) Adenine is always opposite to guanine, cytosine and thymine are opposite to each other in DNA molecule.

Answer: False - Correction: In DNA, adenine is always opposite thymine, and guanine is always opposite cytosine.

(v) DNA molecule is made of two polynucleotide strands.

Answer: True

Exercise - Short questions

(i) Name the carbohydrates suitable as food for man.

The carbohydrates used as food by man are mainly glucose and other simple sugars, starch from cereals, rice, wheat and potatoes, and glycogen from meat. Glucose gives instant energy, while starch is stored food that is broken down to glucose during digestion. Cellulose, though a carbohydrate, cannot be digested by man and only adds roughage.

(ii) Why are fats considered as high energy compounds?

Fats are high energy compounds because their molecules contain many carbon-hydrogen bonds and very little oxygen, so complete oxidation releases much more energy per gram than carbohydrates or proteins: about 9 kilocalories per gram, against about 4 for carbohydrates. That is why the body stores its reserve energy as fat.

(iii) What is the function of mRNA?

Messenger RNA (mRNA) carries the genetic message from DNA in the nucleus to the ribosomes in the cytoplasm. It is copied from one gene of DNA during transcription, and its sequence of bases (codons) tells the ribosome the exact order of amino acids for building a protein. In this way mRNA links the gene to the protein it codes for.

(iv) What is the general formula for amino acids?

The general formula of an amino acid is NH2-CHR-COOH: a central alpha carbon atom joined to an amino group (NH2), a carboxyl group (COOH), a hydrogen atom, and a variable side group (R) which differs in each amino acid and gives it its particular character. Twenty types of amino acids, differing in their R group, build all proteins.

(v) What is the percentage of water in brain cells of man?

The percentage of water in the brain cells of man is 85 percent, as given in the chapter. Water content differs from tissue to tissue: it is very high in brain cells and blood, and much lower in bones and dry seeds. Almost all the reactions of a cell take place in the presence of water.

Exercise - Extensive questions

(i) Describe the importance of water for life.

Water is essential for life for many reasons: it is the solvent in which most chemical reactions of the cell take place, and it carries dissolved food, gases and wastes. Its high heat capacity keeps body temperature steady, and its high heat of vaporisation cools the body by sweating. Water takes part in reactions such as hydrolysis and photosynthesis, gives turgidity to plant cells, and provides the medium in which aquatic organisms live.

(ii) Describe what do you know about polysaccharides.

Polysaccharides are large carbohydrates formed by joining hundreds or thousands of monosaccharides, usually glucose, by glycosidic bonds. The main examples are starch (the storage carbohydrate of plants), glycogen (the storage carbohydrate of animals), cellulose (the structural material of plant cell walls, and the most abundant carbohydrate in nature) and chitin (found in fungal walls and insect skeletons). They are insoluble in water and serve for storage or for structure.

(iii) Write a short note on amino acids.

Amino acids are the building blocks of proteins. Each has an amino group, a carboxyl group, a hydrogen atom and a variable R group joined to a central carbon atom. Twenty types occur in proteins; some are essential and must come from food, while others the body can make. Amino acids join by peptide bonds into long chains that fold into proteins, which build the body, act as enzymes and hormones, and carry out almost every function of the cell.

Chapter 3: Enzymes (Exercise p. 67)

Exercise - Fill in the blanks

(i) Enzymes are composed of hundreds of _____.

Answer: **amino acids**

(ii) Some enzymes consist of a non-protein part known as a _____.

Answer: **cofactor**

(iii) Many enzymes require a non-protein component called _____ for their proper functioning.

Answer: **coenzyme**

(iv) Enzymes are highly _____ in nature.

Answer: **specific**

(v) The enzymes which carry out the synthesis of _____ are integral parts of ribosomes.

Answer: **proteins**

Exercise - Write whether the statement is true or false (correct it if false)

(i) The enzymes important in photosynthesis are found in the mitochondria.

Answer: False - Correction: The enzymes of photosynthesis are found in the chloroplasts. Mitochondria hold the enzymes of respiration.

(ii) Large amounts of an enzyme can accelerate chemical reactions.

Answer: False - Correction: Enzymes work in small amounts and are not used up. Adding more enzyme increases the rate only until all the substrate is being used; beyond that, extra enzyme has little effect.

(iii) Calvin Malvin proposed Lock and Key model for enzyme action.

Answer: False - Correction: The Lock and Key model was proposed by Emil Fischer. Melvin Calvin is known for working out the dark reactions (Calvin cycle) of photosynthesis.

(iv) The active site of an enzyme is composed of four regions.

Answer: False - Correction: The active site is the small region of the enzyme where the substrate binds and the reaction takes place; it is not composed of four regions.

(v) Structure of an enzyme has no specific importance.

Answer: False - Correction: The structure of an enzyme, especially the shape of its active site, is of the greatest importance: it decides which substrate the enzyme fits and therefore which reaction it can speed up.

Exercise - Short questions

(i) List two conditions that destroy enzymatic activity by disrupting bonds between the atoms in an enzyme.

The two conditions are: (1) High temperature: heat breaks the weak bonds that hold the enzyme's shape, so the enzyme is denatured and its active site is lost. (2) Extreme pH: a very acidic or very alkaline medium changes the charges on the enzyme's amino acids, breaks its bonds, and likewise denatures it. In both cases the enzyme can no longer fit its substrate.

(ii) How do low and high temperature affect an enzyme activity?

At low temperature, the molecules move slowly, collisions between enzyme and substrate are rare, and enzyme activity is low, though the enzyme is not destroyed. As temperature rises, activity increases up to an optimum (about 37 C in man). Above the optimum, high temperature breaks the enzyme's shape (denaturation) and activity falls sharply, finally to zero.

(iii) What is a prosthetic group?

A prosthetic group is a non-protein part that is firmly and permanently attached to an enzyme and is needed for the enzyme to work. It may be a metal ion or an organic molecule. Together the protein part (apoenzyme) and the prosthetic group form the complete, active enzyme (holoenzyme). An example is the iron-containing haem group of catalase.

(iv) Define inhibitors of enzyme.

Inhibitors are substances that slow down or stop the working of an enzyme. Competitive inhibitors resemble the substrate and occupy the active site, so the substrate cannot bind. Non-competitive inhibitors bind to another part of the enzyme and change its

shape, so the active site no longer fits the substrate. Some medicines and poisons, such as cyanide, act as enzyme inhibitors.

(v) How does an enzyme accelerate a metabolic reaction?

An enzyme accelerates a reaction by lowering its activation energy: the energy needed to start the reaction. The substrate fits into the enzyme's active site, forming an enzyme-substrate complex, which strains the substrate's bonds and helps them break or join. The products are then released and the enzyme is free to work again, unchanged. In this way a reaction that would be very slow becomes fast.

Exercise - Extensive questions

(i) Describe in detail the mechanism of enzyme action.

The mechanism of enzyme action is explained by two models. In the Lock and Key model (Fischer), the substrate fits exactly into the active site, like a key into a lock, and the enzyme-substrate complex forms. In the Induced Fit model (Koshland), the active site is flexible and changes its shape slightly to fit the substrate snugly. In both models the complex lowers the activation energy, the reaction takes place, products are released, and the enzyme remains unchanged and ready for the next substrate.

(ii) Give the effect of pH and temperature on the efficiency of an enzyme action.

Each enzyme works best at a particular optimum pH (for example pepsin at pH 2 in the stomach, and most enzymes near pH 7) and at an optimum temperature (about 37 C in man). Activity rises towards the optimum as molecular movement increases, but beyond it activity falls: extreme pH changes the enzyme's charges and breaks its bonds, and high temperature denatures it. Low temperature only slows the enzyme without destroying it.

(iii) Write a note on inhibitors of enzymes.

Inhibitors reduce enzyme activity. Competitive inhibitors compete with the substrate for the active site; their effect can be overcome by adding more substrate. Non-competitive inhibitors bind elsewhere on the enzyme, change its shape, and cannot be overcome by more substrate. Irreversible inhibitors bind permanently and destroy the enzyme.

Inhibitors are important in medicine (many drugs are inhibitors) and in the control of metabolism by feedback, where the end product of a pathway inhibits its first enzyme.

(iv) What is the importance of enzymes in life?

Enzymes are essential for life because they make the chemical reactions of the body fast

enough to support life: without them digestion, respiration, photosynthesis and the building of new molecules would be far too slow. They work in small amounts, are not used up, and are highly specific, so each reaction is controlled separately. Enzymes are also used in industry, for example in making bread, cheese, detergents and medicines.

Chapter 4: The Cell (Exercise pp. 101-102)

Exercise - Fill in the blanks

(i) In eukaryotic cell chromatin material is bounded by ----------.

Answer: **the nuclear membrane (nuclear envelope)**

(ii) A group of ribosomes attached to mRNA is known as ----------.

Answer: **a polysome (polyribosome)**

(iii) ---------- is the place on chromosomes where spindle fibres are attached during cell division.

Answer: **The centromere**

(iv) The endoplasmic reticulum with attached ribosomes is known as ----------.

Answer: **rough endoplasmic reticulum**

(v) The soluble part of the cytoplasm is called.

Answer: **cytosol**

Exercise - Write whether the statement is true or false (correct it if false)

(i) Cell membrane is present in all eukaryotic cells.

Answer: True

(ii) Chloroplast and mitochondria do not have hereditary material.

Answer: False - Correction: Both chloroplasts and mitochondria contain their own DNA and ribosomes, and can make some of their own proteins.

(iii) Centriole is involved in cell secretions.

Answer: False - Correction: The Golgi apparatus is involved in cell secretions. Centrioles take part in cell division by organising the spindle.

(iv) Sometimes many ribosomes get attached to the same stretch of mRNA forming a structure called the cytosome.

Answer: False - Correction: Many ribosomes on the same mRNA form a polysome (polyribosome), not a cytosome.

(v) Mitochondria are very important organelles of the eukaryotic cells.

Answer: True

Exercise - Short questions

(i) Describe various movements involved in the transport of materials across the cell membrane.

Materials cross the cell membrane by: diffusion (movement from higher to lower concentration); osmosis (diffusion of water through a semi-permeable membrane); facilitated diffusion (movement through carrier or channel proteins, without energy); active transport (movement against the concentration gradient, using ATP); endocytosis, including phagocytosis (taking in solids) and pinocytosis (taking in liquids); and exocytosis (expelling materials in vesicles).

(ii) State various structural modifications in a cell involved in secretions.

Cells specialised for secretion have a well-developed rough endoplasmic reticulum for making proteins, a large Golgi apparatus for packing and modifying them, many secretory vesicles that carry the product to the membrane, and many mitochondria to supply the energy needed. In gland cells the membrane area may also be increased by folds, and microvilli increase the surface for absorption and release.

(iii) List the processes blocked by mitochondrial failure in a cell.

Mitochondrial failure blocks all the energy-demanding processes of the cell, because ATP production stops. These include active transport across membranes, muscle contraction, nerve conduction, synthesis of proteins and other molecules, cell division, and the working of cilia and flagella. Without ATP the cell cannot maintain itself and dies.

(iv) What will happen if a chromosome loses its centromere?

If a chromosome loses its centromere, the spindle fibres cannot attach to it during cell division. The chromosome will therefore not be pulled to either pole, will be left behind, and will be lost from the daughter cells. The daughter cells will then lack the genes carried on that chromosome, which is usually fatal to the cell.

(v) How does autophagy help in converting a tadpole larva into an adult amphibian?

During metamorphosis the tadpole's tail and other larval structures are no longer needed. Lysosomes carry out autophagy: they digest the cells of the tail and break them down into simple materials. These materials are then reused to build the adult structures, such as legs and lungs. In this way autophagy removes the larval body and recycles its materials for the adult.

(vi) Is there any similarity between bacterial and plant cell wall?

Yes. Both bacterial and plant cell walls are rigid layers outside the cell membrane that give the cell its shape, protect it, and prevent it from bursting when water enters. They differ in composition: the bacterial wall is made of peptidoglycan (murein), while the plant wall is made of cellulose, with pectin and other materials.

Exercise - Extensive questions

(i) Compare structure and function of chloroplasts and mitochondria.

Both chloroplasts and mitochondria are double-membraned organelles with their own DNA and ribosomes, and both make energy available to the cell. Chloroplasts occur only in plant cells (and algae): they contain chlorophyll in thylakoids, trap light energy, and carry out photosynthesis, storing energy in glucose. Mitochondria occur in plant and animal cells: their inner membrane is folded into cristae, and they carry out aerobic respiration, releasing the energy of glucose as ATP. In short, chloroplasts store energy, mitochondria release it.

(ii) State 'Cell Theory' and discuss its emergence.

Cell Theory states: (1) All living things are made of one or more cells. (2) The cell is the basic structural and functional unit of life. (3) All cells arise from pre-existing cells. It emerged after the microscope was improved: Hooke (1665) first saw 'cells' in cork, Leeuwenhoek saw living microbes, Schleiden (1838) said plants are made of cells, Schwann (1839) said the same for animals, and Virchow (1855) added that cells come only from other cells.

(iii) Write notes on: (a) Cytoskeleton (b) Peroxisomes and Glyoxysomes.

(a) Cytoskeleton: the network of protein fibres (microtubules, microfilaments and intermediate filaments) that gives the cell its shape, holds organelles in place, and brings about movement, including cell division and the beating of cilia. (b) Peroxisomes: small

organelles that break down hydrogen peroxide (a harmful by-product) into water and oxygen, and oxidise fatty acids. Glyoxysomes: special peroxysomes of plant seeds that convert stored fats into sugars during germination.

(iv) What might happen if some lysosomal enzymes are absent? Explain with examples.

If a lysosomal enzyme is absent, the material it should digest cannot be broken down and accumulates inside the lysosomes, which swell and damage the cell. This causes storage diseases. For example, in Tay-Sachs disease the enzyme for breaking down a brain lipid is missing, so the lipid accumulates in nerve cells, causing mental retardation and early death. In Pompe's disease the enzyme for glycogen breakdown is missing, so glycogen accumulates in muscles.

Chapter 5: Variety of Life (Exercise p. 122)

Exercise - Fill in the blanks

(i) C. Linnaeus divided all known forms of life into two kingdoms: _____ and ____. Bacteria were placed in the kingdom _____ because they have cell walls, and protozoa were placed in the kingdom _____ because they move from place to place and ingest food.

Answer: **Plantae and Animalia; Plantae; Animalia**

(ii) The most common system of classification used today, developed in 1969 by Robert Whittaker of Cornell University, uses five kingdoms: _____ and _____.

Answer: **Monera, Protista, Fungi, Plantae and Animalia**

(iii) Whittaker's five kingdom system of classification recognizes two basic types of cells: _____ and _____.

Answer: **prokaryotic and eukaryotic**

(iv) In five kingdom system of classification proposed by Margulis and Schwartz, organelles of symbiotic origin such as _____ and _____ were also considered.

Answer: **mitochondria and chloroplasts**

(v) A bacteriophage reproduces by using the metabolic machinery of _____.

Answer: **its host bacterium**

(vi) The protein coat that encloses the viral genome is called: _____ It is made up of

_____.

Answer: **capsid; capsomeres (protein subunits)**

(vii) Retroviruses are _____ viruses which have specific enzymes _____ by which they convert RNA to DNA.

Answer: **RNA; reverse transcriptase**

(viii) HIV infects _____ and the defects in these cells lead to failure in _____ system.

Answer: **helper T cells (T4 lymphocytes); the immune system**

(ix) Hepatitis is caused by _____.

Answer: **virus (hepatitis viruses)**

(x) Viral Hepatitis is of _____ types. Hepatitis A and C are caused by _____ virus whereas _____ virus is the causative agent of Hepatitis B.

Answer: **five; RNA; DNA**

Key notes (this chapter):

- In this source copy, the Chapter 5 exercise contains only the fill-in-the-blanks section (p. 122); no further sections are printed for this chapter before Chapter 6 begins.

Chapter 6: Kingdom Prokaryotae (Monera) (Exercise pp. 142-144)

Exercise - Fill in the blanks

(i) A bacterial arrangement in packets of eight cells is described as a ----------.

Answer: **sarcina**

(ii) The shape and arrangement of _____ is diplococci.

Answer: **cocci in pairs**

(iii) Pili are tubular shafts in bacteria that serve as a means of ----------.

Answer: **attachment and conjugation**

(iv) ---------- are unusual type of bacteria that live in extreme habitats.

Answer: **Archaebacteria**

(v) ---------- is a bacterium that is photosynthetic.

Answer: **A cyanobacterium, e.g. Nostoc**

(vi) ---------- is a cyanobacterium.

Answer: **Nostoc**

(vii) ---------- called as bloom forming organism.

Answer: **Nostoc (and other cyanobacteria, e.g. Anabaena)**

(viii) Use of antibiotics is one of the means of controlling ---------- diseases.

Answer: **bacterial**

Exercise - Short questions

(i) (a) Name general characteristics that could be used to define the prokaryotes. (b) Do any other microbial groups besides bacteria have prokaryotic cells? (c) In what habitats are bacteria found? Give some general means by which bacteria derive nutrients.

(a) Prokaryotes are single-celled organisms without a true nucleus: their DNA lies free in the cytoplasm, they have no membrane-bound organelles, their ribosomes are 70S, and most have a cell wall of peptidoglycan. (b) Yes: the archaebacteria and the cyanobacteria also have prokaryotic cells. (c) Bacteria are found almost everywhere: in soil, water, air, hot springs, salty lakes, and inside other organisms. They obtain nutrients as autotrophs (photosynthesis or chemosynthesis) or as heterotrophs (absorbing dead matter or living as parasites).

(ii) (a) List functions that the cell membrane performs in bacteria. (b) What are mesosomes and some of their possible functions?

(a) The bacterial cell membrane controls what enters and leaves the cell, holds the enzymes of respiration and photosynthesis, helps in the transport of nutrients, and gives a site for DNA attachment during division. (b) Mesosomes are folded inward extensions of the cell membrane. Their possible functions are: increasing the surface for respiration, helping in cell wall formation, and helping in the separation of DNA during cell division.

(iii) What is unique about the structure of bacterial ribosomes?

Bacterial ribosomes are 70S ribosomes, smaller than the 80S ribosomes of eukaryotic cells. Each is made of a large 50S subunit and a small 30S subunit. Because they differ from the host's ribosomes, many antibiotics, such as streptomycin, can block bacterial ribosomes and stop bacterial protein synthesis without harming the patient's cells.

(iv) Draw the three bacterial shapes.

Practical task: draw and label the three shapes: (1) Coccus: a round, spherical cell. (2) Bacillus: a rod-shaped cell. (3) Spirillum (spiral): a curved or spiral cell, including the comma-shaped vibrio and the thin, flexible spirochete. The drawing should show each shape singly and, where asked, in its usual arrangement (pairs, chains or clusters).

(v) Name a bacterium that has no cell wall.

Mycoplasma is a bacterium that has no cell wall. It is the smallest known free-living cell, and because it lacks a rigid wall it has a variable shape and is not affected by antibiotics such as penicillin, which act on the cell wall.

(vi) A gram stained discharge from an abscess shows cocci in irregular grape like clusters. What is the most likely genus of this bacterium?

The most likely genus is Staphylococcus. Staphylococci divide in several planes and the cells remain together in irregular, grape-like clusters. Staphylococcus aureus is a common cause of abscesses, boils and wound infections.

(vii) Draw an outline and label (i) streptobacilli, (ii) diplococci, (iii) staphylococci.

Practical task: draw and label: (i) Streptobacilli: rod-shaped cells joined end to end in chains. (ii) Diplococci: round cells arranged in pairs. (iii) Staphylococci: round cells arranged in irregular, grape-like clusters. Each drawing should show the cell shape and the arrangement that gives the group its name.

(viii) You observe a culture of predominantly round (presumably spherical) bacteria that though apparently fully divided, nevertheless have failed to separate, thus resulting in long chains of cells. What, generally, might you call such an arrangement?

Such an arrangement is called streptococci: round (coccus) cells that divide in one plane and remain joined in long chains. Streptococcus is a well-known example, including the bacteria that cause sore throat.

(ix) Match the following descriptions with the best answer. (a) Division in one plane; cocci arranged in pairs (b) Division in one plane; cocci arranged in chains (c) Division in two planes; cocci arranged in a square of four (d) Division in one plane; rods completely separate after division (e) Division in one plane; rods arranged in chains (f) A comma shaped bacterium (g) A thin, flexible spiral (h) A thick, rigid spiral.

(a) Diplococcus. (b) Streptococcus. (c) Tetrad. (d) Bacillus. (e) Streptobacillus. (f) Vibrio. (g) Spirochete. (h) Spirillum. (Staphylococcus and Sarcina are not used in this matching.)

Exercise - Extensive questions

(i) Describe in detail the structure of bacterial cell wall, emphasizing Gram positive and Gram negative properties.

The bacterial cell wall is a rigid layer outside the cell membrane, made mainly of peptidoglycan (murein): chains of sugars cross-linked by short peptides. In Gram-positive bacteria the wall is thick, with many layers of peptidoglycan and teichoic acids; it keeps the violet Gram stain and is not damaged by the decolouriser. In Gram-negative bacteria the wall is thin, with a single peptidoglycan layer plus an outer membrane of lipopolysaccharides; it loses the violet stain and takes the red counter-stain. The wall gives shape, protects the cell, and prevents bursting.

(ii) Write an account of different methods used for controlling microbes.

Microbes are controlled by: physical methods, such as heat (boiling, autoclaving, pasteurisation), cold, drying, filtration and ultraviolet radiation; chemical methods, such as disinfectants (phenol, alcohol, chlorine) for surfaces and antiseptics for skin; antibiotics, which kill or stop the growth of bacteria inside the body; and immunisation, which protects by vaccination. Clean water, proper sewage disposal, food hygiene and quarantine also control the spread of microbes.

(iii) Discuss the role of antibiotics and immunization in controlling bacterial diseases. What problem can arise due to misuse of antibiotics?

Antibiotics are drugs that kill bacteria or stop their growth, curing diseases such as pneumonia, tuberculosis and cholera. Immunisation (vaccination) prevents bacterial diseases, such as diphtheria, tetanus and whooping cough, by training the immune system before infection. Misuse of antibiotics, such as stopping a course early or using them for viral diseases, lets resistant bacteria survive and multiply, producing antibiotic-resistant strains (such as MRSA) that are very hard to treat.

(iv) Describe general characteristics of Cyanobacteria with special reference to Nostoc.

Cyanobacteria are photosynthetic prokaryotes, formerly called blue-green algae. They have chlorophyll a and carry out photosynthesis with release of oxygen, they have no true nucleus or chloroplasts, and many can fix atmospheric nitrogen. Nostoc is a common example: it forms bead-like chains (trichomes) of round cells in a jelly, with larger heterocysts that fix nitrogen. Nostoc lives in ponds, moist soil and rice fields, forms water

blooms, and improves soil fertility by nitrogen fixation.

(v) Write notes on: (a) Koch's postulates (b) Shape of bacteria (c) Flagella and pili (d) Growth in bacteria.

(a) Koch's postulates: the rules for proving that a germ causes a disease: the germ must be found in every case of the disease, isolated in pure culture, able to cause the same disease when given to a healthy host, and recovered again from that host. (b) Shape of bacteria: coccus (round), bacillus (rod) and spiral (vibrio, spirillum, spirochete), with arrangements such as pairs, chains, clusters and packets. (c) Flagella and pili: flagella are long whip-like threads for movement; pili are short, thin shafts for attachment and for conjugation. (d) Growth in bacteria: bacteria grow by binary fission, doubling at regular intervals; growth passes through lag, log (exponential), stationary and death phases.

Key notes (this chapter):

- Chapter 6, Q.1 has two items numbered (vii) in print, and Q.1 (ii) is garbled in print ('The shape and arrangement of is diplococci'); it is given here as 'The shape and arrangement of cocci in pairs is diplococci.' Chapter 6, Q.2 (ix) is a matching question whose answer list is lettered (a)-(j) against descriptions (a)-(h); the correct pairs are given in the answer.

Chapter 7: The Kingdom Protista (Exercise p. 171)

Exercise - Short questions

(i) Write two characteristics of each of the following groups: (i) Protozoa (ii) Dinoflagellates (iii) Diatoms (iv) Slime molds (v) Oomycetes.

(i) Protozoa: single-celled, animal-like protists; they move (by pseudopodia, cilia or flagella) and ingest food, and most have no cell wall. (ii) Dinoflagellates: single-celled algae with two flagella and a covering of cellulose plates; many are photosynthetic and some cause red tides. (iii) Diatoms: single-celled algae with a glassy cell wall of silica, made of two overlapping halves like a box; they are a major source of food in water. (iv) Slime molds: fungus-like protists that creep as a plasmodium and form spore-bearing structures for reproduction. (v) Oomycetes: fungus-like protists (water molds) with

filamentous bodies; they differ from fungi by having cellulose walls and diploid bodies.

Exercise - Extensive questions

(i) Discuss important features of protists. Why are protists so difficult to classify?

Protists are mostly single-celled eukaryotes, but they show great variety: some are plant-like and photosynthesise, some are animal-like and ingest food, and some are fungus-like. They live mainly in water and moist places, and reproduce asexually and sometimes sexually. They are difficult to classify because they do not fit neatly into plants, animals or fungi: one group may show characters of two kingdoms, and different scientists group them differently. That is why they are kept in their own kingdom, Protista.

(ii) What are the reasons for grouping simple eukaryotic organisms into a separate kingdom, Protista?

Simple eukaryotes are grouped in Protista because they are eukaryotic yet mostly single-celled, unlike the many-celled plants, animals and fungi. They show a mixture of plant-like, animal-like and fungus-like characters, so placing them in any one of the other kingdoms would be wrong. A separate kingdom recognises their distinct level of organisation and their great diversity, and it keeps the other kingdoms natural.

(iii) How are protists important to humans? What is their ecological importance?

Protists are important to humans because photosynthetic protists (algae, diatoms, dinoflagellates) produce much of the world's oxygen and form the base of aquatic food chains that support fisheries. Diatoms are used in filters and polishes, and some protists cause diseases such as malaria and sleeping sickness. Ecologically, protists are producers, consumers and decomposers in water: they feed small animals, recycle nutrients, and keep aquatic ecosystems running.

(iv) What are three major groups of protists?

The three major groups of protists are: (1) Algae: photosynthetic, plant-like protists, such as diatoms, dinoflagellates and green algae. (2) Protozoa: animal-like, heterotrophic protists, such as Amoeba, Paramecium and Plasmodium. (3) Fungus-like protists: slime molds and oomycetes (water molds), which resemble fungi in body form and reproduction but differ in cell structure.

(v) Discuss general characteristics of algae.

Algae are photosynthetic protists (and some are many-celled) that contain chlorophyll and make their own food. They live mainly in water and moist places, have simple bodies without true roots, stems or leaves, and their cell walls are usually made of cellulose. They range from single cells (diatoms) to large seaweeds, reproduce asexually and sexually, and form the base of most aquatic food chains.

(vi) Green algae are considered ancestral organisms of green land plants. Discuss.

Green algae are considered the ancestors of green land plants because they share the same pigments (chlorophyll a and b), store food as starch, and have cellulose cell walls, all of which land plants also have. Their chloroplasts are similar in structure, and some green algae show a simple alternation of generations like land plants. These shared characters show that land plants evolved from a green algal ancestor that moved onto land.

(vii) What features distinguish Oomycetes from fungi?

Oomycetes (water molds) differ from true fungi in several features: their cell walls are made of cellulose, not chitin; their bodies are diploid for most of the life cycle, while fungi are haploid or dikaryotic; they produce motile, flagellated spores (zoospores), which fungi do not; and their sexual reproduction forms oospores. These differences show that oomycetes are protists, not fungi, although they look fungus-like.

(viii) Describe structure and reproduction of slime molds.

A slime mold has two stages. In the feeding stage it is a plasmodium: a creeping, multinucleate mass of cytoplasm without a cell wall, which moves like an amoeba and engulfs food particles. In the reproductive stage it forms upright, spore-bearing structures (sporangia); the spores are released, germinate, and fuse to form a new plasmodium. In cellular slime molds the cells stay separate but gather together to form a slug-like body that then makes spores.

Key notes (this chapter):

- Chapter 7, Q.1 lists Oomycetes under a second (iv) in print, and Q.2 jumps from (iv) to (vi); the questions are given here in their printed order with the numbering tidied.

Chapter 8: Fungi - The Kingdom of Recyclers (Exercise pp. 193-194)

Exercise - Short questions

(i) What is a hypha? What is the advantage of having incomplete septa?

A hypha is a single thread-like filament of a fungus; a mass of hyphae is called a mycelium. In many fungi the hyphae are divided by cross walls (septa), but the septa are incomplete, with pores in the middle. The advantage is that cytoplasm, nutrients and even nuclei can flow freely from cell to cell along the hypha, so food absorbed at one end can quickly reach the growing tip.

(ii) What is the composition of fungal cell wall and how is this composition advantageous to fungi?

The fungal cell wall is made mainly of chitin (with glucans and proteins), a tough, nitrogen-containing polysaccharide. Chitin gives the wall great strength and rigidity, so it protects the delicate hyphae, keeps the cell's shape, and prevents bursting when water enters. It also resists decay, which suits fungi that live on dead matter, and it is not digested by most animals, so fungi are well protected.

(iii) To which phyla do yeasts belong? How do they differ from other fungi?

Yeasts belong to the phylum Ascomycota (sac fungi). They differ from most other fungi by being single-celled instead of forming a mycelium of hyphae, and by reproducing mainly by budding, in which a small outgrowth forms on the parent cell and separates. Most yeasts do not form the large fruiting bodies seen in mushrooms and molds.

(iv) Name sexual and asexual spores of Ascomycetes.

In Ascomycetes the sexual spores are ascospores: usually eight spores formed inside a sac called an ascus, after the fusion of nuclei and meiosis. The asexual spores are conidia: spores cut off from the tips of special hyphae (conidiophores), which spread the fungus quickly without sexual reproduction.

(v) What are mycorrhizae?

Mycorrhizae are close associations between the hyphae of a fungus and the roots of a plant, in which both partners benefit. The fungus greatly increases the root's absorbing surface and supplies the plant with water and minerals, especially phosphorus, while the

plant supplies the fungus with sugars made in photosynthesis. Most land plants have mycorrhizae.

(vi) By what means can individuals in imperfect fungi be classified?

Imperfect fungi (Deuteromycota) have no known sexual stage, so they cannot be classified by their sexual structures. They are classified instead by their asexual characters: the form of their mycelium, the type and shape of their asexual spores (conidia), and the way the conidia are produced. When a sexual stage is later found, the fungus is moved to its true phylum.

(vii) Give a single characteristic that differentiates Zygomycota from Basidiomycota.

Zygomycota form thick-walled resting spores called zygospores by the fusion of two similar gametangia, and their hyphae are usually without septa. Basidiomycota form sexual spores called basidiospores on club-shaped basidia, and their hyphae are septate and often dikaryotic. The single clear difference is: zygospore in Zygomycota, basidiospore on a basidium in Basidiomycota.

(viii) Why is green mold more likely to contaminate an orange kept in a refrigerator than are bacteria?

Green mold (Penicillium) is more likely to contaminate a refrigerated orange because fungi tolerate cold, dry and acidic conditions much better than most bacteria. The orange peel is acidic and fairly dry, and the refrigerator's low temperature slows bacteria strongly but allows the mold to grow slowly. Also, mold spores are common in the air and settle easily on fruit.

(ix) What is a fungus?

A fungus is a eukaryotic, heterotrophic organism that obtains food by absorption: it secretes enzymes onto dead matter or a living host and absorbs the digested nutrients. Its body is usually a mycelium of thread-like hyphae, its cell walls are made of chitin, and it reproduces by spores. Fungi are not plants, because they have no chlorophyll and cannot photosynthesise.

(x) State two parallel characteristics of Ascomycetes and Basidiomycetes.

Two parallel characteristics are: (1) Both have septate hyphae, divided by cross walls with pores. (2) Both have a dikaryotic stage in their life cycle, in which cells contain two separate nuclei (n + n) before the nuclei fuse and meiosis forms the sexual spores (ascospores in Ascomycetes, basidiospores in Basidiomycetes).

Exercise - Extensive questions

(i) Discuss taxonomic status of fungi.

Fungi were once placed with plants because they are fixed and have cell walls, but they are now given their own kingdom, Fungi. This is because they differ from plants in fundamental ways: they have no chlorophyll and cannot photosynthesise, their walls are made of chitin (not cellulose), they store glycogen (not starch), and they feed by absorption. Molecular evidence also shows that fungi are more closely related to animals than to plants. They are therefore a separate kingdom of heterotrophic eukaryotes.

(ii) Summarise differentiating/distinguishing characteristics of four main groups of Fungi, and give two common examples of each group.

(1) Zygomycota: hyphae without septa; sexual spores are zygospores. Examples: Rhizopus (bread mold), Mucor. (2) Ascomycota: sexual spores (ascospores) formed inside sacs (asci); yeasts are included. Examples: Saccharomyces (yeast), Penicillium. (3) Basidiomycota: sexual spores (basidiospores) formed on clubs (basidia); includes mushrooms. Examples: Agaricus (mushroom), Puccinia (wheat rust). (4) Deuteromycota (imperfect fungi): no known sexual stage; classified by asexual spores. Examples: Aspergillus, Alternaria.

(iii) State various features of fungi that adapt them to terrestrial mode of life.

Fungi are adapted to land by: tough chitin walls that prevent drying and damage; hyphae that grow through soil and dead matter to absorb food; spores with resistant walls that survive dry conditions and spread through air; the ability to digest complex dead matter outside the body; and mycorrhizal associations with plant roots, which help both partners obtain water and minerals on land.

(iv) What is ecological importance of saprotrophic fungi, of lichens and mycorrhizae?

Saprotrophic fungi are the great recyclers: they break down dead plants and animals, returning carbon, nitrogen and minerals to the soil for new growth; without them the world would be buried in dead matter. Lichens (a fungus plus an alga) are pioneer organisms that colonise bare rock, break it down, and start soil formation. Mycorrhizae help most land plants absorb water and minerals, especially phosphorus, and so support forests and crops.

(v) Some enzymes of fungi are useful on one hand and harmful on other. Discuss.

Fungal enzymes are useful because they are used in industry: to make bread, beer and cheese, to produce antibiotics such as penicillin, and to soften and flavour food. The same enzymes are harmful when fungi attack living things: they digest the tissues of crops (rusts, smuts, blights), rot stored food and timber, and break down the skin, nails and lungs of humans, causing diseases such as ringworm and aspergillosis. The same power of digestion thus helps and harms.

(vi) Name any four important fungal diseases of plants and four fungal diseases of humans, and briefly describe any one of the plant diseases and any one of the diseases of humans.

Plant diseases: wheat rust (Puccinia), rice blast, potato late blight, and powdery mildew of grapes. Human diseases: ringworm, athlete's foot, candidiasis (thrush), and aspergillosis. Wheat rust: the fungus Puccinia forms orange pustules on wheat leaves and stems, steals the plant's food, and can destroy the whole crop. Ringworm: a fungus of the skin (dermatophyte) grows in the outer skin, causing itchy, ring-shaped, red patches on the skin, scalp or nails.

(vii) Describe, giving examples, different ways in which fungi are useful to humans.

Fungi are useful in many ways: yeasts (Saccharomyces) ferment sugar to make bread, beer and wine; molds (Penicillium) give cheese its flavour and produce the antibiotic penicillin; mushrooms (Agaricus) are eaten as food; fungi produce industrial enzymes and organic acids (citric acid); mycorrhizal fungi help crops grow; and saprotrophic fungi recycle waste and clean the environment by breaking down pollutants.

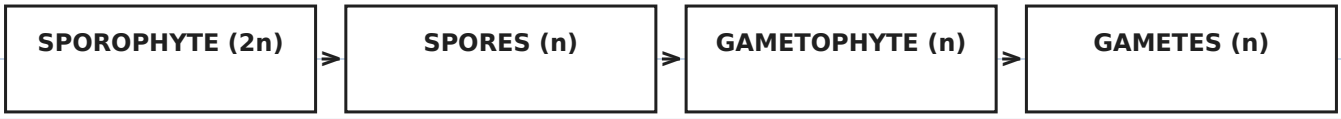
(viii) Differentiate between the members of each of the following pairs. (a) Spore/Conidium (b) Ascus/Basidium (c) Dikaryotic/Diploid (d) Ascocarp/Ascus (e) Obligate parasite/Facultative parasite (f) Endomycorrhizae/Ectomycorrhizae (g) Plasmogamy/Karyogamy.

(a) Spore is any reproductive cell that can grow into a new fungus; a conidium is a special asexual spore cut off from a hypha. (b) An ascus is a sac in which ascospores form; a basidium is a club on which basidiospores form. (c) Dikaryotic means a cell has two separate nuclei (n + n); diploid means one nucleus with two sets of chromosomes (2n). (d) An ascocarp is the fruiting body containing asci; an ascus is the single sac inside it. (e) An obligate parasite can live only on a living host; a facultative parasite usually lives on dead matter but can attack a living host. (f) In endomycorrhizae the hyphae enter the root cells; in ectomycorrhizae they stay outside, forming a sheath around the root. (g)

Plasmogamy is the fusion of cytoplasm (bringing two nuclei together); karyogamy is the later fusion of those two nuclei.

Chapter 9: Kingdom Plantae (Exercise pp. 239-241)

Alternation of generations



Exercise - Fill in the blanks

(i) The sporophyte is _____ and _____ generation and the gametophyte is _____ and _____.

Answer: **diploid (2n) and asexual; haploid (n) and sexual**

(ii) The motile asexual reproductive cells are characteristics of _____ and are called _____.

Answer: **algae (and lower plants); zoospores**

(iii) The sexual reproduction is said to be oogamous or heterogamous if the two fusing gametes are ----------.

Answer: **different in size and form (a large non-motile egg and a small motile sperm)**

(iv) In the stem of Monocotyledons the bundles are _____ while in the stem of Dicotyledons they are _____.

Answer: **scattered; arranged in a ring**

(v) The double fertilization is the characteristic feature of _____.

Answer: **angiosperms (flowering plants)**

(vi) Stem, roots and leaves are the _____ parts and flowers, fruits and seeds are the _____ parts of the plant.

Answer: **vegetative; reproductive**

(vii) _____ is the phenomenon of the production of two kinds of spores in the plants.

Answer: **Heterospory**

(viii) The naked-seeded plants are included in the group _____.

Answer: **Gymnosperms**

Exercise - Short questions

(i) (a) How are ferns better adapted to life on land than liverworts and mosses? (b) Which of the following are nutritionally self supporting: 1. Mature liverwort and moss gametophyte. 2. Mature liverwort and moss sporophyte.

(a) Ferns are better adapted to land because they have true roots, stems and leaves, and a vascular system (xylem and phloem) for carrying water and food, so they can grow tall. Their sporophyte is the large, independent plant, while in liverworts and mosses the sporophyte is small and depends on the gametophyte. Ferns also have a better-protected sporangium and stronger tissues. (b) The mature liverwort and moss gametophyte is nutritionally self-supporting (it is green and photosynthetic); the sporophyte is not, because it depends on the gametophyte for food.

(ii) (a) The chances of survival and development of wind-blown pollen grains are much less than those of spores of Adiantum. Comment on this statement. (b) Account for the fact that megaspores are large and microspores are small. (c) What important advances have angiosperms made towards the seed plant life?

(a) The statement is true in the sense that a pollen grain must land on the right stigma of the same species to succeed, and most are lost, while an Adiantum spore needs only moist soil to germinate into a gametophyte. But pollen is protected and carried in huge numbers, and its success rate is raised by pollination mechanisms. (b) Megaspores are large because they store food for the female gametophyte and the future embryo; microspores are small because they only need to produce a few male gametes and must be made in huge numbers for wind dispersal. (c) Angiosperms advanced by enclosing the seed in a fruit, by double fertilization forming a nutritive endosperm, by flowers that attract pollinators, and by faster, more efficient vessels.

(iii) Write a note on the alternation of generations.

Alternation of generations is the regular alternation, in the life cycle of a plant, between two generations: the sporophyte and the gametophyte. The diploid sporophyte produces haploid spores by meiosis; each spore grows into a haploid gametophyte, which produces gametes; the gametes fuse to form a diploid zygote, which grows into a new sporophyte. In bryophytes the gametophyte is dominant, while in ferns and seed plants the sporophyte is dominant.

(iv) What is the importance of the following? (i) Seed (ii) Double fertilization (iii) Heterospory.

(i) Seed: it protects the embryo, supplies it with stored food, and lets it survive drought and spread to new places, so the plant no longer needs water for fertilisation. (ii) Double fertilization: one sperm fuses with the egg to form the embryo, and the second fuses with the central cell to form the triploid endosperm, which feeds the embryo; food is therefore made only when fertilisation has succeeded. (iii) Heterospory: producing two kinds of spores (microspores and megaspores) was the step that led to the seed habit, because the female gametophyte could then be protected and nourished on the parent plant.

(v) Pick and match the following.

Fern sporophyte: the diploid generation. Moss plant: the gametophytic generation. Gamete: the last cell of the gametophyte. Spore: the first cell of the gametophyte. Spores: haploid, asexual reproductive cells. Vegetative reproduction: involves the vegetative parts of plants. Oospore: the first cell of the sporophyte. Spore mother cell: divides by reduction division (meiosis) to form haploid spores. Gymnosperms: the plants with naked seeds. (The printed matching list is scrambled, with duplicated and broken entries; the pairs above are the correct matches.)

(vi) Sketch and label a fertile pinnule and a sporangium of Adiantum.

Practical task: draw a fertile pinnule (leaflet) of the fern Adiantum (maidenhair fern), showing the leaflet blade with its edge folded back to protect the sporangia (false indusium), and a single sporangium: a small capsule on a stalk, with a ring of thick-walled cells (annulus) that throws the spores out when it dries. Label: pinnule, folded margin (indusium), sporangium, stalk, annulus, and spores.

Exercise - Extensive questions

(i) To what does alternation of generations refer in the plants? Define sporophyte and gametophyte. With which stage is an adult animal comparable? How are they reproductively dissimilar?

Alternation of generations refers to the alternation of a diploid spore-producing generation with a haploid gamete-producing generation. The sporophyte is the diploid (2n) generation that produces spores by meiosis. The gametophyte is the haploid (n) generation that produces gametes by mitosis. An adult animal is comparable to the

sporophyte, because it is diploid. They differ reproductively because the animal produces gametes directly by meiosis, while the plant sporophyte produces spores by meiosis and the gametes are produced later by the gametophyte.

(ii) What is a seed? Why is the seed a crucial adaptation to terrestrial life?

A seed is a fertilised ovule containing an embryo, stored food, and a protective seed coat. It is a crucial adaptation to land because it protects the embryo from drying and damage, feeds it until it can photosynthesise, lets it wait (dormancy) for good conditions, and spreads the plant to new places. Because of the seed, plants no longer need water for fertilisation and can live in dry places far from water.

(iii) Describe evolution of leaf and its importance in vascular plants.

Leaves are thought to have evolved in two ways: microphylls (small leaves with a single vein, as in club mosses) from small outgrowths of the stem, and megaphylls (large leaves with branching veins, as in ferns and seed plants) from flattened, webbed branch systems. Leaves are important because they are the main photosynthetic organs: their broad, flat blades catch light, their stomata control gas exchange, and their veins carry water in and food out, letting vascular plants grow large and productive.

(iv) Discuss evolution of seed and its significance.

The seed evolved from heterospory: plants began to make two kinds of spores, and the megaspore was kept on the parent plant instead of being shed. The megasporangium became surrounded by protective coats (integuments), forming the ovule, and after fertilisation the ovule became the seed. Its significance is that the embryo is protected, nourished and dispersed, and fertilisation no longer needs external water, so seed plants could conquer dry land and become the dominant plants of the earth.

(v) In what way do the flowering plants differ from the rest of the seed plants? What is the stigma? Is fertilization in angiosperms direct or indirect? From what tissue does angiosperm fruit develop?

Flowering plants (angiosperms) differ from other seed plants (gymnosperms) by having flowers, by enclosing their seeds inside an ovary that becomes a fruit, and by double fertilization. The stigma is the sticky top of the carpel that receives pollen. Fertilisation in angiosperms is indirect: the pollen tube grows down the style and delivers the sperms to the ovule inside the ovary, rather than swimming to the egg. The fruit develops from the wall of the ovary (and sometimes nearby tissues).

(vi) What two classes comprise the angiosperms? How do the two classes structurally differ from one another? Which class derived from the other? Explain.

The two classes are Monocotyledonae (monocots) and Dicotyledonae (dicots). They differ in: seed (one cotyledon in monocots, two in dicots); leaves (parallel veins in monocots, net veins in dicots); stem bundles (scattered in monocots, in a ring in dicots); roots (fibrous in monocots, tap root in dicots); and flower parts (in threes in monocots, in fours or fives in dicots). Dicots are considered the older group, and monocots are believed to have derived from early dicots.

Key notes (this chapter):

- Chapter 9, Q.1 has no item (viii) in print, and Q.2 has no item (v) in print; the questions are given in their printed order.

Chapter 10: Kingdom Animalia (Exercise pp. 290-292)

Exercise - Fill in the blanks

(i) Protozoans have been placed in a separate kingdom known as _____.

Answer: **Protista**

(ii) The sponges do not have any symmetry and are therefore called _____.

Answer: **asymmetrical**

(iii) Between ectoderm and endoderm the coelenterates have a non cellular _____.

Answer: **mesoglea**

(iv) Taenia solium has _____ and _____ for attachment to the intestine of host.

Answer: **hooks and suckers**

(v) In annelids the body segmentation of the type known as _____.

Answer: **metameric segmentation**

(vi) In insects there are _____ pairs of legs present in the _____ region of the body.

Answer: **three; thoracic**

(vii) The organ of locomotion in molluscs is the _____.

Answer: **foot**

(viii) In animals where there are definite left and right sides the symmetry is _____.

Answer: **bilateral**

(ix) The system in which water moves inside the body of an echinoderm is called _____ .

Answer: **the water vascular system**

(x) Coelom is the body cavity formed from the _____ layer.

Answer: **mesoderm**

Exercise - Choose the correct option

(i) Vertebrates that develop embryonic membranes around their embryo are called:

a) Amniotes ✓

b) Anamniotes

(ii) In animals the bodies of which can be divided in two equal halves only in one plane are:

a) asymmetrical

b) bilaterally symmetrical ✓

c) radially symmetrical

(iii) Animals that have their body cavity filled with parenchyma are:

a) Acoelomates ✓

b) Coelomates

c) Pseudocoelomates

(iv) The vertebrates in which placenta is formed during the development of foetus are:

a) Pisces

b) Aves

c) Mammalia ✓

(v) In amphibians the necessary requirements to spend their life history are:

a) land

b) water

c) both ✓

(vi) Trypanosoma causes the disease:

a) Malaria

b) Sleeping sickness ✓

(vii) In annelids the organs of excretion are:

a) flame cells

b) nephridia ✓

c) kidneys

(viii) In arthropoda the body cavity is:

a) pseudocoelom

b) enterocoel

c) haemocoel ✓

(ix) In mollusca the foot is used for:

a) capturing prey

b) locomotion ✓

c) both

Exercise - Extensive questions

(i) What are Cnidaria? Explain the diploblastic origin, alternation of generations in Cnidaria.

Cnidaria (coelenterates) are aquatic, mostly marine animals with radial symmetry, a sac-like body with one opening, and stinging cells (cnidoblasts) on their tentacles; examples are Hydra, jellyfish and corals. They are diploblastic: their body wall has only two layers, an outer ectoderm and an inner endoderm, with a non-cellular mesoglea between them. They show alternation of generations between two forms: the fixed, asexual polyp and the free-swimming, sexual medusa, as in Obelia.

(ii) Describe the parasitic adaptations in phylum Platyhelminthes. How does tapeworm affect a person?

Parasitic flatworms are adapted to life inside a host by: a body covering (tegument) that resists the host's digestive juices; hooks and suckers for attachment (Taenia has both); loss of digestive system, absorbing digested food through the body wall; a flattened body for absorption; and enormous egg production to ensure transmission. A tapeworm in a person absorbs digested food from the intestine, causing weakness, weight loss, abdominal pain, anaemia and, in heavy infections, blockage of the intestine.

(iii) Give the symptoms of the disease caused by certain nematodes.

Nematode diseases show symptoms according to the worm: Ascaris (roundworm) causes abdominal pain, malnutrition, blockage of the intestine, and coughing when larvae pass

through the lungs. Hookworms cause anaemia and weakness by sucking blood from the intestine. Pinworms cause itching around the anus, especially at night. Filarial worms block the lymph vessels, causing swelling of the limbs (elephantiasis).

(iv) Give an account of the major groups of Arthropods. What is the economic importance of insects?

The major groups of arthropods are: Crustacea (crabs, prawns, mostly aquatic, with two pairs of antennae); Myriapoda (centipedes and millipedes, with many legs); Arachnida (spiders, scorpions, mites, with four pairs of legs and no antennae); and Insecta (the largest group, with three body parts, three pairs of legs and usually wings). Insects are economically important because bees pollinate crops and make honey, silkworms give silk, and lac insects give lac; but many insects are pests of crops and carriers of disease, such as mosquitoes and locusts.

(v) Give the two major classes of the Pisces and explain the adaptations of aquatic mode of life in fishes.

The two major classes of Pisces are Chondrichthyes (cartilaginous fishes, such as sharks and rays) and Osteichthyes (bony fishes, such as rohu and trout). Their aquatic adaptations are: a streamlined body for swimming; fins for movement and balance; gills for breathing dissolved oxygen; a lateral line system for sensing vibrations; a swim bladder (in bony fishes) for buoyancy; and scales and mucus that reduce friction and protect the body.

(vi) Give the adaptations for aerial mode of life in birds. What is their origin?

Birds are adapted for flight by: forelimbs modified into wings with feathers; light, hollow bones; a strong keeled breastbone for flight muscles; a highly efficient respiratory system with air sacs; a high metabolic rate and warm blood; a short, light body with no teeth and no urinary bladder; and acute eyesight. Birds are believed to have originated from small, feathered, bipedal reptiles (theropod dinosaurs); Archaeopteryx is the famous fossil link between reptiles and birds.

(vii) What are the general characteristics of mammals? How do the three subclasses Prototheria, Metatheria and Eutheria differ from one another?

Mammals are warm-blooded vertebrates with hair on the body, mammary glands that produce milk, a four-chambered heart, a diaphragm, and a well-developed brain; most give birth to living young. The subclasses differ in reproduction: Prototheria (monotremes,

such as duck-billed platypus) lay eggs. Metatheria (marsupials, such as kangaroo) give birth to very immature young that complete development in a pouch. Eutheria (placental mammals, such as man) nourish the embryo through a placenta and give birth to well-developed young.

(viii) Distinguish between the following by giving examples: (a) Radial and Bilateral Symmetry (b) Diploblastic and Triploblastic animals (c) Anamniotes and Amniotes.

(a) Radial symmetry: the body can be divided into similar halves by several planes through the centre, as in Hydra and starfish. Bilateral symmetry: the body can be divided into right and left halves by only one plane, as in man, frog and earthworm. (b) Diploblastic animals have two germ layers (ectoderm and endoderm), as in Hydra; triploblastic animals have three (ectoderm, mesoderm, endoderm), as in earthworm and man. (c) Anamniotes have no embryonic membranes and lay eggs in water, as in fishes and amphibians; amniotes have embryonic membranes (amnion) and lay shelled eggs or bear live young, as in reptiles, birds and mammals.

Key notes (this chapter):

- Chapter 10, Q.2 has an unnumbered lead item (Amniotes/Anamniotes) before items (i)-(vii), and item (vi) combines two questions (annelid excretory organs; arthropod body cavity), which are presented here as two separate items.

Chapter 11: Bioenergetics (Exercise pp. 326-327)

Exercise - Write whether the statement is true or false (correct it if false)

(i) Hydroponics are the plants grown in water culture.

Answer: True

(ii) Calcium is an essential element for chlorophyll formation.

Answer: False - Correction: Magnesium, not calcium, is essential for chlorophyll formation: a magnesium atom sits at the centre of the chlorophyll molecule. Calcium is needed for cell walls and membranes.

(iii) Chlorosis means yellowing of leaves due to deficiency of certain essential element of plant nutrition.

Answer: True

Exercise - Short questions

(i) List four features of a leaf which show that it is able to carry out photosynthesis effectively.

Four features are: (1) A large, flat blade that catches maximum sunlight. (2) A thin structure, so gases diffuse quickly to all cells. (3) Many chloroplasts in the mesophyll cells, rich in chlorophyll. (4) Stomata on the surface for the exchange of carbon dioxide and oxygen, and veins that bring water in and carry food away.

(ii) Summarise the role of water in photosynthesis.

Water is a raw material of photosynthesis: in the light reactions it is split (photolysis), releasing oxygen, electrons and protons. The electrons replace those lost by chlorophyll, the protons help build ATP, and the oxygen is released to the air. Water also keeps the leaf cells turgid, carries minerals, and provides the hydrogen that finally reduces carbon dioxide to glucose.

(iii) What are T.W. Engelmann and Melvin Calvin famous for?

T.W. Engelmann is famous for showing, with the alga Spirogyra and oxygen-seeking bacteria, that photosynthesis is strongest in red and blue light: the action spectrum of photosynthesis. Melvin Calvin is famous for tracing the path of carbon in photosynthesis using radioactive carbon-14, and discovering the cycle of reactions (the Calvin cycle) by which carbon dioxide is fixed into sugars.

(iv) What is the difference between an action spectrum and an absorption spectrum?

An absorption spectrum shows how much light of each wavelength a pigment (such as chlorophyll) absorbs. An action spectrum shows how effective each wavelength is in causing photosynthesis, measured by the rate of oxygen release or carbon dioxide uptake. The two are similar but not identical, because accessory pigments also absorb light and pass the energy to chlorophyll.

(v) What is the role of accessory pigments in light absorption?

Accessory pigments (chlorophyll b, carotenoids and xanthophylls) absorb wavelengths of light that chlorophyll a absorbs poorly, especially blue and green light, and pass that

energy to chlorophyll a at the reaction centre. They widen the range of light the plant can use, and carotenoids also protect chlorophyll from damage by excess light.

(vi) When and why is there no net exchange of CO2 and O2 between the leaves and the atmosphere?

There is no net exchange at the compensation point, at low light intensity (usually at dawn and dusk). At that light level the rate of photosynthesis exactly equals the rate of respiration: the carbon dioxide released by respiration is used up by photosynthesis, and the oxygen released by photosynthesis is used up by respiration, so no net gas exchange is seen.

(vii) What is the net production of ATP during glycolysis?

The net production of ATP during glycolysis is 2 ATP molecules per glucose molecule. Glycolysis makes 4 ATP in total, but 2 ATP are used up in the early steps to activate glucose, so the net gain is 2 ATP, together with 2 NADH.

(viii) What is the main difference between photophosphorylation and oxidative phosphorylation?

Photophosphorylation makes ATP using light energy: it takes place in the chloroplasts during photosynthesis, where light drives electrons and the energy released is used to add phosphate to ADP. Oxidative phosphorylation makes ATP using chemical energy: it takes place in the mitochondria during respiration, where electrons from NADH and FADH2 pass down the electron transport chain and the energy released is used to make ATP.

(ix) What is the location of ETC and chemiosmosis in photosynthesis and cellular respiration?

In photosynthesis the electron transport chain and chemiosmosis are located on the thylakoid membranes of the chloroplast: protons are pumped into the thylakoid space and flow back through ATP synthase. In cellular respiration they are located on the inner mitochondrial membrane (cristae): protons are pumped into the intermembrane space and flow back through ATP synthase to make ATP.

(x) How did the evolution of photosynthesis affect the metabolic pathway?

The evolution of photosynthesis changed life's metabolism completely: it released free oxygen into the atmosphere, which allowed aerobic respiration to evolve, giving far more energy (36 ATP per glucose) than anaerobic pathways. It also made organic food

available to all heterotrophs, built the ozone layer that protects life from ultraviolet rays, and created the food chains on which almost all life depends.

(xi) How does absorption spectrum of chlorophyll a differ from that of chlorophyll b?

Chlorophyll a absorbs most strongly in blue-violet light (about 430 nm) and in red light (about 660 nm). Chlorophyll b absorbs most strongly in blue light (about 450 nm) and in orange-red light (about 640 nm): its peaks are shifted slightly from those of chlorophyll a. Together they absorb a wider range of light than either alone.

(xii) Why are the carotenoids usually not obvious in the leaves? They can be seen in the leaves before leaf fall. Why?

Carotenoids are usually not obvious because their yellow-orange colour is masked by the much larger amount of green chlorophyll in the leaf. Before leaf fall, in autumn, chlorophyll breaks down and is withdrawn from the leaf, while the more stable carotenoids remain, so their yellow and orange colours become visible and the leaves turn yellow and orange.

(xiii) How is the formation of vitamin A linked with eating of carrot?

Carrots are rich in beta-carotene, a carotenoid pigment. When we eat carrots, the beta-carotene is absorbed in the intestine and converted in the liver into vitamin A (retinol). Vitamin A is needed for good vision (it forms part of the visual pigment rhodopsin), for healthy skin, and for growth, so eating carrots helps prevent vitamin A deficiency.

Exercise - Extensive questions

(i) Explain the roles of the following in aerobic respiration: (a) NAD+ and FAD (b) oxygen.

(a) NAD+ and FAD are hydrogen carriers (coenzymes): during glycolysis and the Krebs cycle they accept hydrogen atoms removed from glucose and its products, becoming NADH and FADH2. They then carry the hydrogen to the electron transport chain, where the electrons are passed down and the energy is used to make ATP, and NAD+ and FAD are freed to be used again. (b) Oxygen is the final electron acceptor of the electron transport chain: it accepts electrons and hydrogen to form water. Without oxygen the chain stops, NADH cannot be recycled, and aerobic respiration cannot continue.

(ii) Sum up how much energy (as ATP) is made available to the cell from a single glucose molecule by the operation of glycolysis, the formation of acetyl CoA, the citric acid cycle,

and the electron transport chain.

By the classic textbook accounting: glycolysis gives a net 2 ATP and 2 NADH; the formation of acetyl CoA gives 2 NADH; the citric acid (Krebs) cycle gives 2 ATP and 6 NADH plus 2 FADH2. When the NADH and FADH2 pass through the electron transport chain, they yield about 32 ATP more. The gross total is 38 ATP, of which 2 ATP are spent transporting NADH into the mitochondria, leaving a net gain of 36 ATP per glucose molecule for the cell.

(iii) Trace the fate of hydrogen atoms removed from glucose during glycolysis when oxygen is present in muscle cells; compare this to the fate of hydrogen atoms removed from glucose when the amount of the available oxygen is insufficient to support aerobic respiration.

When oxygen is present, the hydrogen atoms removed during glycolysis are carried by NADH into the mitochondria, passed down the electron transport chain, and finally combined with oxygen to form water, releasing energy for ATP. When oxygen is insufficient, the electron transport chain cannot run, so the hydrogen is passed to pyruvate instead: in muscle cells pyruvate is reduced to lactic acid, which regenerates NAD+ so that glycolysis can continue, but yields only 2 ATP per glucose and causes muscle fatigue.

(iv) Sketch Krebs cycle and discuss its energy yielding steps.

Practical task: draw the Krebs (citric acid) cycle as a circle: acetyl CoA (2C) joins oxaloacetate (4C) to form citrate (6C), which is then oxidised step by step back to oxaloacetate, releasing two molecules of carbon dioxide. The energy-yielding steps are those that produce NADH (three per turn: at isocitrate, alpha-ketoglutarate and malate steps), FADH2 (one, at the succinate step) and one ATP (GTP) at the succinyl CoA step. Since two acetyl CoA enter per glucose, the cycle yields 6 NADH, 2 FADH2 and 2 ATP in total.

(v) Describe various steps involved in oxidative break down of glucose to pyruvate.

The oxidative breakdown of glucose to pyruvate is glycolysis, which takes place in the cytoplasm in two phases. In the energy-investment phase, glucose is phosphorylated using 2 ATP and split into two molecules of glyceraldehyde-3-phosphate. In the energy-yielding phase, each of these is oxidised: NAD+ accepts hydrogen to form NADH, and phosphate is added to make ATP by substrate-level phosphorylation. The end product

is two molecules of pyruvate, with a net gain of 2 ATP and 2 NADH per glucose.

(vi) Sketch respiratory electron transport chain. Discuss the significance of ETC.

Practical task: draw the inner mitochondrial membrane with its chain of carriers (NADH dehydrogenase, cytochromes, and ATP synthase): electrons from NADH and FADH2 pass from carrier to carrier, releasing energy that pumps protons into the intermembrane space; the protons flow back through ATP synthase, making ATP; oxygen accepts the electrons at the end to form water. The significance of the ETC is that it produces most of the cell's ATP (about 32 of the 36 ATP per glucose), recycles NAD+ and FAD, and uses oxygen safely to make water.

(vii) Compare photosynthesis with respiration in plants.

Photosynthesis and respiration are opposite but linked processes. Photosynthesis takes place in chloroplasts, uses light energy, carbon dioxide and water, and produces glucose and oxygen; it stores energy. Respiration takes place in mitochondria, uses glucose and oxygen, and produces carbon dioxide, water and energy (ATP); it releases energy. Photosynthesis occurs only in light, respiration goes on all the time. In a green plant the two balance each other, so the plant both makes and uses its own food.

(viii) Explain the difference between the cyclic and non-cyclic photophosphorylation with the help of Z scheme.

In non-cyclic photophosphorylation, electrons flow in a Z-shaped path: light excites photosystem II, electrons pass down a chain to photosystem I, are excited again, and finally reduce NADP+ to NADPH; water is split to replace the electrons, releasing oxygen, and ATP is made as well. In cyclic photophosphorylation, only photosystem I is used: the excited electrons return to photosystem I through the chain, making ATP but no NADPH and no oxygen. Non-cyclic photophosphorylation thus gives ATP, NADPH and oxygen; cyclic gives only ATP.

(ix) Give an account of light-independent reactions of photosynthesis.

The light-independent reactions (the Calvin cycle) take place in the stroma of the chloroplast and do not need light directly, but use the ATP and NADPH made in the light reactions. Carbon dioxide is fixed by the enzyme RuBisCO onto ribulose bisphosphate (RuBP) to form a 6-carbon compound that splits into two molecules of PGA. Using ATP and NADPH, PGA is reduced to PGAL (triose phosphate); some PGAL leaves the cycle to make glucose and other sugars, while the rest uses ATP to regenerate RuBP, so the cycle

continues.

Chapter 12: Nutrition (Exercise pp. 361-363)

Exercise - Fill in the blanks

(i) Plants absorb minerals in their _____ form, as found in the soil.

Answer: **ionic**

(ii) In plants the most common nutrient deficiencies are of _____, _____ and _____.

Answer: **nitrogen, phosphorus and potassium (NPK)**

(iii) A plant requires _____ for holding its cell together.

Answer: **calcium**

(iv) Most of the organic material in a plant is _____.

Answer: **carbohydrate (mainly cellulose)**

(v) Chlorosis is usually caused by insufficient _____ .

Answer: **magnesium**

(vi) In _____ the trapped insects are decomposed by bacteria.

Answer: **the pitcher plant (Sarracenia)**

(vii) The structure in the mouth that prevents food from entering the nasal cavities is the _____ .

Answer: **soft palate**

(viii) The stomach functions to _____ food and its _____ .

Answer: **churn; digestion**

(ix) _____ is the common example of detritivore.

Answer: **Earthworm**

(x) Pancreas produces _____ which stimulates the conversion of glycogen to _____ .

Answer: **glucagon; glucose**

(xi) Vomiting occurs due to _____ movements.

Answer: **antiperistaltic**

Exercise - Short questions

(i) What is the advantage of a digestive tract as compared with a digestive cavity?

A digestive tract is a complete tube with a mouth and an anus, so food moves in one direction and digestion takes place in stages: different regions specialise in ingestion, digestion, absorption and egestion. In a digestive cavity (a sac with one opening, as in Hydra) food must enter and leave by the same opening, so digestion cannot be staged and undigested food mixes with fresh food. The tract is therefore more efficient and allows continuous feeding.

(ii) What are functions of human liver?

The liver's functions are: it produces bile, which emulsifies fats; it stores glucose as glycogen and releases it when needed; it converts excess amino acids to urea (deamination); it detoxifies drugs and poisons; it stores iron and vitamins A and D; it makes blood plasma proteins such as fibrinogen; and it destroys old red blood cells. It is the body's largest chemical factory.

(iii) What measures should be taken to avoid food poisoning?

To avoid food poisoning: wash hands and utensils before handling food; wash fruits and vegetables; cook food thoroughly, especially meat; keep cooked and raw food separate; store perishable food in a refrigerator; do not eat stale or expired food; keep food covered from flies; and drink clean, boiled water. These measures prevent germs from reaching and growing in food.

(iv) Can we get along without large intestine? If not, why?

We can survive without part of the large intestine, but not well without all of it, because the large intestine has important jobs: it absorbs most of the remaining water and salts from the undigested food, so the faeces become solid; its bacteria make some vitamins (such as vitamin K and some B vitamins); and it stores and expels the faeces. Without it, water and salts would be lost and the body would suffer dehydration.

Exercise - Extensive questions

(i) Define nutrition. Describe the role of different elements in plant nutrition.

Nutrition is the process by which an organism takes in food and uses it for growth, energy and repair. In plant nutrition, the major elements are: nitrogen (for proteins and

chlorophyll), phosphorus (for ATP, DNA and membranes), potassium (for enzyme activity and stomatal opening), calcium (for cell walls and membranes), magnesium (the centre of chlorophyll), sulphur (for some amino acids), and iron (for chlorophyll formation and enzymes). Lack of any one causes deficiency diseases such as chlorosis and stunted growth.

(ii) (a) Distinguish between saprophytic and parasitic modes of life. (b) (i) Name one parasitic plant. (ii) Describe its method of nutrition, explaining why normal nutrition is not possible.

(a) Saprophytes feed on dead and decaying organic matter, digesting it outside the body and absorbing the products; parasites feed on a living host, taking food from it and usually harming it. Saprophytes are useful recyclers; parasites depend on the host's life.

(b) (i) Cuscuta (dodder) is a parasitic plant. (ii) Cuscuta has no proper leaves or chlorophyll, so it cannot photosynthesise; it twines around a host plant and sends suckers (haustoria) into the host's tissues to draw ready-made food and water from it. Normal nutrition is impossible because it lacks the green pigment and leaves needed for photosynthesis.

(iii) What are heterotrophs? Describe different methods of nutrition in heterotrophs.

Heterotrophs are organisms that cannot make their own food and must obtain ready-made organic food from other organisms. Their methods of nutrition are: holozoic (taking in solid food and digesting it inside the body, as in man and Amoeba); saprophytic (feeding on dead matter, as in fungi); parasitic (feeding on a living host, as in tapeworm and Cuscuta); symbiotic (two organisms living together and sharing food, as in lichens); and insectivorous (catching insects for nitrogen, as in the pitcher plant).

(iv) What are the advantages and disadvantages of the parasitic mode of life compared with that of a free living organism?

Advantages of parasitism: food and shelter are ready-made and constant, so the parasite needs no food-searching structures; reproduction is often very high to ensure transmission. Disadvantages: the parasite depends completely on the host and dies if the host dies; it faces the host's immune defences; it must have special adaptations for attachment and transmission; and its body often degenerates, losing organs such as the digestive system, so it can never live freely again.

(v) Why is digestion necessary? Describe what happens to a meal containing fats,

carbohydrates and proteins while it is in the stomach of man.

Digestion is necessary because food is taken in as large, insoluble molecules that cannot pass through cell membranes or be used by cells; digestion breaks them into small, soluble molecules (glucose, amino acids, fatty acids) that can be absorbed and used. In the stomach, the meal is churned and mixed with gastric juice: hydrochloric acid kills germs and activates pepsin, which begins digesting proteins into peptides; mucus protects the stomach wall; fats and carbohydrates are hardly digested here (only slight fat digestion by gastric lipase), and the meal becomes a semi-liquid chyme that passes to the small intestine.

(vi) What is holozoic nutrition? Describe the characteristic processes involved in holozoic nutrition giving the example of Amoeba.

Holozoic nutrition is the taking in of solid food particles, digesting them inside the body, and absorbing the products, as in animals. In Amoeba the processes are: ingestion, by surrounding the food particle with pseudopodia to form a food vacuole; digestion, by enzymes secreted into the vacuole; absorption, of the digested food into the cytoplasm; assimilation, using the food for energy and growth; and egestion, expelling the undigested remains through the cell surface. These five steps are characteristic of holozoic nutrition.

(vii) How do (i) the saliva, (ii) the pancreas, (iii) the liver help in the digestion of the food of man? Draw a diagram of the digestive system to show the positions of the pancreas and the liver.

(i) Saliva moistens food, contains mucus for swallowing, and has salivary amylase (ptyalin) that begins digesting starch into maltose. (ii) The pancreas secretes pancreatic juice with amylase (starch), lipase (fats) and trypsin (proteins), and bicarbonate to neutralise the stomach acid. (iii) The liver secretes bile, which emulsifies fats into small droplets so that lipase can digest them, though bile itself has no enzyme. Practical task: draw the human digestive system and label the liver (upper right abdomen) and the pancreas (behind the stomach).

(viii) Make a labelled diagram of the alimentary canal and digestive glands in cockroach. What are the functions of the glands you sketch?

Practical task: draw the cockroach's alimentary canal: mouth, pharynx, oesophagus, crop (for storage), gizzard (for grinding), midgut or mesenteron (digestion and absorption),

hindgut (ileum, colon, rectum) and anus; and the glands: salivary glands (secrete saliva to moisten food and digest starch), hepatic caecae at the junction of gizzard and midgut (secrete digestive enzymes), and Malpighian tubules (excretion, not digestion). The glands digest food and the tubules remove nitrogenous waste.

(ix) Describe the structure and functions of human stomach.

The human stomach is a J-shaped, muscular bag between the oesophagus and the small intestine, with four regions: cardiac (where food enters), fundus, body and pylorus (where food leaves). Its wall has three muscle layers that churn the food, and glands that secrete gastric juice: hydrochloric acid, pepsinogen (activated to pepsin for protein digestion), mucus (protecting the wall) and rennin in infants. Its functions are to store food, churn it, begin protein digestion, kill germs with acid, and pass chyme to the duodenum.

(x) How do the digestive tracts of herbivores differ from those of carnivores?

Herbivores have longer digestive tracts than carnivores, because plant food (cellulose) is hard to digest and needs more time and a larger surface. Herbivores often have a large caecum or a multi-chambered stomach (as in ruminants) where bacteria ferment cellulose. Carnivores have shorter tracts, a larger, simpler stomach with strong acid for digesting meat, and a small caecum, because meat is easy to digest and rich in protein.

(xi) What prevents the wall of stomach from being digested?

The stomach wall is protected from its own acid and pepsin by: a thick layer of alkaline mucus secreted by the stomach lining, which coats the wall and neutralises acid; the tight junctions between the lining cells, which stop acid leaking through; the rapid replacement of the lining cells (every few days); and the fact that pepsinogen is only activated to pepsin after it leaves the cells, in the stomach cavity. If the mucus layer breaks down, an ulcer forms.

(xii) What specialized features of your small intestine account for the efficient absorption of digested foodstuffs?

The small intestine absorbs efficiently because: it is very long (about 6 metres), giving a large surface; its inner wall is folded and covered with finger-like villi, and each villus cell has microvilli, increasing the surface enormously; the villi are richly supplied with blood capillaries and lacteals, so absorbed food is carried away quickly; the wall is thin (one cell thick), so diffusion is fast; and the slow movement of food gives time for complete absorption.

(xiii) What is the contribution of liver and pancreas in the process of digestion?

The liver contributes bile, which is stored in the gall bladder and released into the duodenum: bile salts emulsify fats into tiny droplets, greatly increasing the surface for lipase, and bile also neutralises stomach acid. The pancreas contributes pancreatic juice, containing amylase for starch, lipase for fats, trypsin and chymotrypsin for proteins, and bicarbonate to neutralise acid. Together they make the digestion of all three food types possible in the small intestine.

(xiv) How can we control obesity?

Obesity can be controlled by: eating a balanced diet with fewer fats and sugars and more vegetables, fruits and fibre; controlling portion sizes and avoiding overeating; taking regular exercise, such as walking, running or games, to burn extra calories; avoiding soft drinks and junk food; eating regular meals instead of snacking; and, in severe cases, seeking a doctor's advice. The key is to use as much energy as is taken in.

(xv) How is gastric juice production regulated?

Gastric juice production is regulated in three phases. In the cephalic phase, the sight, smell or thought of food sends nerve signals (vagus nerve) that start acid secretion before food arrives. In the gastric phase, food in the stomach stretches its wall and its proteins stimulate the hormone gastrin, which increases acid and pepsin secretion. In the intestinal phase, when chyme enters the duodenum, hormones such as secretin and the enterogastric reflex slow down stomach secretion and emptying, so digestion is controlled.

Key notes (this chapter):

- Chapter 12, Q.3 numbering is misordered in print: after (xii) comes (xiii), then (xvi), then (xv); the questions are given here in the order (xiii), (xvi), (xv) as printed, under the headings 'contribution of liver and pancreas', 'control of obesity', and 'gastric juice regulation'.

Chapter 13: Gaseous Exchange (Exercise pp. 389-390)

Exercise - Fill in the blanks

(i) _____ is the most abundant protein in the world.

Answer: **RuBisCO (ribulose bisphosphate carboxylase)**

(ii) Haemoglobin is a complex molecule which contains 9512 atoms and _____ amino acids.

Answer: **574**

(iii) The opening of larynx is called _____.

Answer: **glottis**

(iv) When the smaller bronchi attain the diameter of _____ mm or less they are called bronchioles.

Answer: **1**

(v) There are about _____ stomata per square centimeter of leaf surface of tobacco plant.

Answer: **12,000**

Exercise - Write whether the statement is true or false (correct it if false)

(i) ATP is generated during organismic respiration.

Answer: False - Correction: ATP is generated during cellular respiration, not during organismic respiration. Organismic respiration is breathing (ventilation): the exchange of gases between the organism and the environment, which itself produces no ATP.

(ii) Water is a better respiratory medium than air.

Answer: False - Correction: Air is a better respiratory medium than water: air holds far more oxygen, oxygen diffuses much faster in air, and moving air costs less energy than moving water.

(iii) The earthworm does not possess specialized organs for respiration.

Answer: True

(iv) In parabronchi of birds, the blood flows in the opposite direction of air flow.

Answer: True

(v) Ring shaped cartilages are present in trachea of man.

Answer: False - Correction: The cartilages of the human trachea are C-shaped, not complete rings: the open part faces the oesophagus so that food can pass easily.

Exercise - Short questions

(i) How does breathing differ from respiration?

Breathing is the physical process of moving air in and out of the lungs (ventilation): it is simply the exchange of gases between the body and the environment. Respiration is the whole biochemical process by which cells break down glucose with oxygen to release energy (ATP), producing carbon dioxide and water. Breathing is one step that supplies oxygen for respiration; respiration itself takes place inside the cells.

(ii) How much carbon dioxide is present in venous and arterial blood?

Venous blood (returning to the heart and lungs) contains more carbon dioxide, about 53 ml per 100 ml of blood, carried mainly as bicarbonate. Arterial blood (leaving the lungs) contains less carbon dioxide, about 48 ml per 100 ml of blood, because carbon dioxide is released in the lungs during expiration. The difference (about 5 ml per 100 ml) is the carbon dioxide removed in one passage through the lungs.

(iii) How does air always remain in the lungs of human beings?

Air always remains in the lungs because even after a forced expiration, a volume of air (the residual volume, about 1200 ml) stays in the lungs and cannot be expelled. This happens because the lungs are elastic but are held open inside the chest, and the small airways close before all the air can leave. The residual air keeps the alveoli open and lets gas exchange continue between breaths.

(iv) What are the products which are produced during photorespiration?

The products of photorespiration are: carbon dioxide, which is released (so photorespiration wastes fixed carbon); phosphoglycolate, which is processed through the peroxisome and mitochondrion to form glycine and serine, releasing ammonia and another carbon dioxide; and no ATP or NADPH is gained — in fact ATP is used up. Photorespiration is therefore a wasteful process that lowers the efficiency of photosynthesis.

(v) How much denser is a water medium than air medium for exchange of respiratory gases?

Water is about 800 times denser than air, and oxygen diffuses about 10,000 times more slowly in water than in air. Water also holds far less dissolved oxygen than air holds as a gas. That is why aquatic animals must spend much more energy moving water over their

gills, and why air is a far better respiratory medium than water.

Exercise - Extensive questions

(i) In what ways is air a better respiratory medium than water?

Air is better than water because: it contains about 20 times more oxygen per unit volume; oxygen diffuses about 10,000 times faster in air than in water; air is about 800 times less dense, so moving it over a respiratory surface costs far less energy; and air does not need to be pumped continuously in large volumes. Water breathers must therefore work much harder to obtain the same amount of oxygen.

(ii) What is photorespiration? Give its consequences.

Photorespiration is the process in which the enzyme RuBisCO, instead of fixing carbon dioxide, fixes oxygen onto RuBP, producing phosphoglycolate. Its consequences are all harmful to the plant: fixed carbon is lost as carbon dioxide, ATP and reducing power are wasted, ammonia is released and must be re-fixed, and the rate of photosynthesis falls, especially in hot, dry conditions when stomata close and oxygen builds up inside the leaf.

(iii) Describe briefly the properties of respiratory surfaces in cockroach.

The cockroach breathes through a tracheal system: air enters through spiracles on the body wall and passes through tracheae that branch into fine tracheoles reaching every tissue. The respiratory surfaces (the tracheole walls) are thin, moist and close to the tissues, so gases diffuse directly to the cells without blood transport. The spiracles can open and close to reduce water loss, and body movements help pump air in and out.

(iv) In what ways is respiration in birds the most efficient and elaborate?

Bird respiration is the most efficient because: air flows through the parabronchi in one direction only, and blood flows in the opposite direction (counter-current exchange), so oxygen is taken up along the whole length; the lungs are supplemented by air sacs that store air and keep fresh air flowing through the lungs during both inspiration and expiration; and the thin parabronchial walls give a huge surface. Birds therefore extract more oxygen than mammals, which suits the high energy demand of flight.

(v) Discuss the mechanical aspects of breathing in man.

Breathing in man is driven by changes in chest volume. In inspiration, the diaphragm contracts and flattens and the external intercostal muscles raise the ribs, increasing chest volume; air pressure in the lungs falls below atmospheric pressure, and air rushes

in. In expiration, the diaphragm and intercostal muscles relax, the chest volume falls, lung pressure rises above atmospheric pressure, and air is pushed out. Inspiration is active (uses energy), while quiet expiration is passive.

(vi) Write a detailed note on respiratory pigments.

Respiratory pigments are coloured proteins that carry oxygen (and carbon dioxide) in the blood. The main one in man is haemoglobin, found in red blood cells: each molecule has four haem groups with iron, binds four oxygen molecules in the lungs (where oxygen is high) to form oxyhaemoglobin, and releases them in the tissues (where oxygen is low). Other pigments are myoglobin in muscles (stores oxygen), haemocyanin in some molluscs and crustaceans (copper-based, blue when oxygenated), and haemerythrin in some worms. Without pigments, blood could carry far too little oxygen.

(vii) List the air passage way in sequence from nostrils to alveoli. Describe the structure of alveolus in detail.

The air passageway is: nostrils (external nares) -> nasal cavities -> pharynx -> larynx (voice box) -> trachea (windpipe) -> bronchi (one to each lung) -> bronchioles -> alveolar ducts -> alveolar sacs -> alveoli. An alveolus is a tiny, thin-walled air sac, about 0.2 mm across: its wall is a single layer of flat squamous cells, surrounded by a dense network of blood capillaries, with elastic fibres that let it stretch and recoil. The wall is moist, so gases dissolve and diffuse quickly between the air and the blood, and the huge number of alveoli (about 300 million) gives an enormous surface for exchange.

Chapter 14: Transport (Exercise pp. 454-455)

Water pathways in the root



Exercise - Fill in the blanks

(i) In the process of facilitated diffusion, the carrier molecules are _____.

Answer: **specific (each carrier carries only one type of molecule)**

(ii) Pure water has a water potential which is equal to _____.

Answer: **zero**

(iii) The insects which feed on the phloem of plants are the _____.

Answer: **aphids**

(iv) The substance produced by basophils which inhibits blood-clotting is _____.

Answer: **heparin**

(v) The most abundant compound of blood plasma of man is _____.

Answer: **water**

(vi) _____ in 1874 suggested that water molecules move along the cell walls of xylem vessels due to imbibition.

Answer: **Sachs**

Exercise - Write whether the statement is true or false (correct it if false)

(i) The intercellular openings in the blood capillaries are larger than the openings in the lymph capillaries.

Answer: False - Correction: The openings in lymph capillaries are larger than those in blood capillaries: that is how large proteins, bacteria and cell debris can enter the lymph, while blood capillaries keep them out.

(ii) Between the left auricle and the left ventricle in human heart, the valve present is called tricuspid valve.

Answer: False - Correction: The valve between the left auricle (atrium) and the left ventricle is the bicuspid (mitral) valve, with two flaps. The tricuspid valve is on the right side of the heart, between the right auricle and the right ventricle.

(iii) The pacemaker of the heart of man is the AV node.

Answer: False - Correction: The pacemaker of the human heart is the SA node (sinoatrial node) in the wall of the right auricle, which starts each heartbeat. The AV node only relays the impulse to the ventricles after a short delay.

(iv) Each sieve tube member is associated with one or more tracheid cells.

Answer: False - Correction: Each sieve tube member is associated with one or more companion cells, which are living cells that support the sieve tube member (which has

no nucleus). Tracheids are xylem cells for water transport, not phloem companions.

(v) The method of active immunization is used to combat active infections of tetanus and rabies.

Answer: False - Correction: Passive immunization is used to combat an active infection of tetanus or rabies, because ready-made antibodies give immediate protection. Active immunization (vaccination) is preventive: it takes time to build immunity and is given before infection.

Exercise - Extensive questions

(i) How are minerals and water taken up by roots? Draw the structures involved and the pathways for water and minerals from soil water to xylem, and the transport processes at each step.

Water and minerals enter the root mainly through the root hairs, by osmosis (water) and diffusion or active transport (minerals). From the root hair they move through the cortex to the xylem by three pathways: the apoplast pathway (through cell walls and intercellular spaces), the symplast pathway (through the cytoplasm and plasmodesmata), and the vacuolar pathway (through cell vacuoles). At the endodermis the Casparian strip blocks the apoplast pathway, forcing water and minerals through the living cells into the symplast, and from there into the xylem. Practical task: draw the root section and label root hair, epidermis, cortex, endodermis, Casparian strip, pericycle and xylem, with arrows for the three pathways.

(ii) Describe the mechanism of opening and closing of stomata.

Stomata open and close by changes in the turgidity of their two guard cells. In light, the guard cells actively take up potassium ions and produce malate, so their solute concentration rises; water enters by osmosis, they become turgid, and because their inner walls are thicker, they bend outward, opening the pore. In darkness or when water is scarce, potassium leaves the guard cells, water follows out, they become flaccid, and the pore closes. Abscisic acid (ABA) also promotes closure during drought.

(iii) How does the pressure-flow theory explain the movement of sugars through a plant?

The pressure-flow (mass flow) theory explains sugar transport in the phloem as follows: sugars are actively loaded into the sieve tubes at the source (such as a leaf), raising the solute concentration, so water enters by osmosis and builds a high pressure. At the sink

(such as a root or fruit), sugars are unloaded, water leaves, and the pressure falls. The pressure difference between source and sink drives the sugary sap to flow from source to sink through the sieve tubes.

(iv) Describe cohesion-tension theory of water movement in xylem. What supplies the cohesion, and what is the source of tension? How do these two forces interact to move water through plant?

The cohesion-tension theory says that water is pulled up the xylem by transpiration. The source of tension is transpiration from the leaves: water evaporating from mesophyll cells pulls water from the xylem, creating a negative pressure (tension) that travels down the water column. Cohesion is supplied by hydrogen bonds between water molecules, which hold the water column together, and adhesion holds water to the xylem walls. Together, tension pulls from above and cohesion keeps the column unbroken, so water rises continuously from roots to leaves, even in tall trees.

(v) Explain apoplast, symplast and vacuolar pathways, and describe the movement of water and dissolved minerals, through them.

The apoplast pathway is movement through the non-living parts: cell walls and intercellular spaces, without crossing any membrane; it is fast but is blocked by the Casparian strip at the endodermis. The symplast pathway is movement through the living cytoplasm of cells, connected by plasmodesmata; minerals must cross membranes, often by active transport, to enter it. The vacuolar pathway is movement through the vacuoles of cells, crossing the tonoplast at each cell. Water and minerals usually use a mixture of all three until the endodermis, where they are forced into the symplast.

(vi) Explain water potential. What is the relationship of water potential with solute potential and pressure potential?

Water potential is the tendency of water to move from one place to another, measured in megapascals; water always moves from higher to lower water potential. Pure water at atmospheric pressure has a water potential of zero, and adding solutes or applying pressure changes it. The relationship is: water potential = solute potential + pressure potential. Solute potential is always negative (solutes lower water potential), while pressure potential is usually positive in turgid cells (the wall pressing back) and zero in flaccid cells.

(vii) Name and describe the general functions of the three major type of cells or cell like

bodies found in blood of humans. Which of these cell types is found predominantly in lymph?

The three major types are: (1) Red blood cells (erythrocytes): biconcave discs without nuclei, containing haemoglobin, which carry oxygen from the lungs to the tissues and bring back some carbon dioxide. (2) White blood cells (leukocytes): larger, nucleated cells that defend the body by engulfing germs (phagocytes) and by making antibodies (lymphocytes). (3) Platelets (thrombocytes): small cell fragments that help blood to clot and stop bleeding. White blood cells, especially lymphocytes, are found predominantly in lymph.

(viii) Write a note on immunity and its types.

Immunity is the body's ability to resist and destroy disease-causing germs and foreign substances. Its types are: innate (natural) immunity, present from birth, including the skin, mucus, stomach acid, phagocytes and inflammation; and acquired (adaptive) immunity, developed during life. Acquired immunity is of two kinds: active immunity, produced by the body's own response after infection or vaccination, which is long-lasting; and passive immunity, given by ready-made antibodies from the mother or by injection, which is immediate but short-lived. Artificial active immunity is vaccination; artificial passive immunity is antiserum injection.

Key notes (this chapter):

- Chapter 14, Q.3 jumps from (vii) to (ix) in print (no (viii)); the questions are given in their printed order.